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Optimal leader-follower affine formation control of linear multi-agent systems

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Abstract

This article studies optimal leader-follower affine formation control for linear multi-agent systems, which has more formation flexibility due to the usage of stress matrices. First, based on the inverse optimality strategy, an affine formation control law is designed to drive the followers to their target positions, in which a corresponding performance index is minimized. Next, to minimize a prespecified performance index, another class of optimal affine formation control law is proposed by using a cooperative estimator. The design of the cooperative estimator is based on the proportional-integral feedback which estimates the target position of each follower. Under the proposed control law, the followers achieve the target formation and the given performance indices are guaranteed to be optimal at the same time. Finally, the corresponding simulations are carried out to verify the effectiveness of the presented approaches.

KEYWORDS

affine formation control, linear multi-agent systems, optimal control

1 | INTRODUCTION

With the rapid development of modern communication and computer technology, the formation control of multi-agent systems (MASs) has become one of the most actively research topics in the community of systems and control,[1] which has a broad application background such as robotic swarms[2] and unmanned aerial vehicles,[3] and so forth. As the foundation of formation control problem, consensus and containment problems have been extensively investigated. Zheng et al.[4] have proposed consensus protocols for hybrid MASs. Also, Zheng et al.[5] has considered the containment control problem for heterogeneous MASs. As compared to the consensus and containment problems, the formation control has more practical applications. The main existing formation approaches can be divided into position-, displacement-, distance-, bearing, and angle-based formation control according to the sensed and control variables.[6-8] Under different tasks and available information, each approach has its own advantages and disadvantages. For example, displacement-based formation approach has the freedom to the time-varying translation of a target formation,[9] but it is difficult to achieve scaling or rotation. Distance-based control law has freedom to time-varying translation and rotation,[10,11] but the time-varying scaling is not straightforward to be achieved by using this method. Bearing-based control law has the freedom to the translation and scaling of the formation,[12] but the rotation is difficult to be achieved. The position-based control requires each agent to have the common knowledge of a global coordinate frame, for example, GPS receiver. Recently, researchers have proposed some methods which enhance the invariance of the new constraints such as barycentric coordinates,[13] complex Laplacians,[14] stress matrices, and angle constraints.[15] Complex Laplacians are merely applicable to the formation control in the planar case, while the stress matrices-based control method is flexible in dimensions and invariant to any affine transformation with respect to the target formation, such as translation, rotation, scaling, shearing. Therefore, by dividing the MASs into leaders and followers, the formation can be reshaped according to the affine transformation indicated by the leaders.[16] In this article, we adopt this stress matrices-based approach to achieve various affine formations.

Up to now, affine formation control problem has received increasing attention. The affine formation control laws for MASs with single- and double-integrator dynamics and unicycle models are studied in Reference 16. The affine formation of MASs with triple-integrator dynamics is analyzed in Reference 17. Moreover, the work in Reference 18 studies the affine formation control problems with time-varying delays. In addition, proportional-integral control law is proposed in Reference 19 to achieve an affine formation. The affine formation control law with a sliding-mode estimator is also developed in Reference 20. Furthermore, the work in Reference 21 studies the obstacle avoidance control in affine formation. Note that most of the existing results on affine formation control only consider achieving the target formation and do not take the control cost and energy consumption into account.

In practice, each agent in MASs has limited fuel or power such as the gasoline of vehicles and the power of the robot, and so forth. Achieving a desired formation by using the least energy is very important in many engineering situations. Thus, the optimal formation control is of high practical and economic interest, for example, optimal formation tracking problem for planar vehicles.[22] For MASs, the optimal consensus control for leader-following MASs has been studied in some literature. For example, the paper[23] has considered the optimal topology design for leader-following consensus problem. The paper[24] has studied the optimal synchronization control via off-policy reinforcement learning under a directed tree. The optimal containment control algorithm of MASs with strongly connected graph has been proposed in Reference 25, which not only drives the followers states into a convex hull of the leaders' states but also minimizes their transient responses. The papers[26,27] have considered the suboptimal distributed control problem for MASs. To the authors' best knowledge, the optimal control of affine formation problem has not been fully investigated yet, which is significant in reducing energy and fuel consumption during the formation process. Therefore, in this work, we study optimal stress-matrices-based affine formation control problem under an undirected graph. The main contributions of this work can be summarized as follows:

- (i) The optimal stress-matrix-based affine formation control problem is studied for followers in this article. We show that by properly choosing the control gains, the designed optimal control law can guarantee the predesigned cost function minimum. To the authors' best knowledge, the optimal affine formation control based on stress matrices has rarely been investigated in the existing literature.
- (ii) Besides the optimal control law, we also design the other inverse optimal control law, in which the state estimation is not required among agents. Therefore, compared to the optimal control, the inverse optimal control has simpler control structure and does not need the interagent communication.
- (iii) Compared to the existing literature studying the cooperative control of integrator MASs, such as,[16] the agents' dynamics are more general in this article. In addition, the affine formation is more flexible and general than containment configurations since all the followers can be inside or outside of the convex hull spanned by the leaders.

The control task of this article is to drive the followers into the target formation that is affinely localized by leaders and to minimize the performance indices consisting of agents' motion and control cost simultaneously. Optimal control for followers in MASs is complicated due to the fact that the graph topology interplays with system dynamics. Therefore, the tool of inverse optimality is used to obtain the optimal control for MASs.[28] In this article, we have proposed two different optimal control laws. First, the inverse optimal affine formation control law is designed, in which a corresponding performance index is minimized. Next, to minimize a free-selected performance index, the optimal affine formation control algorithm with estimators is proposed. Based on proportional-integral (PI) feedback, the cooperative estimator is designed to estimate the target position of each follower. Then, the performance indices are modified based on the estimated position. Finally, the optimal control law is designed by using PI feedback, which guarantees the new performance indices optimal and the convergence of tracking error simultaneously.

The subsequent sections are organized as follows. Preliminary results and notations used in this article are presented in Section 2. The inverse optimal affine formation control is formulated in Section 3. The optimal affine formation control of the linear MASs is designed in Section 4. Section 5 provides the simulations of presented approaches. Finally, Section 6 gives the conclusion.

2 | NOTATION AND PRELIMINARIES

2.1 | Notation

The superscripts "T" and " − 1" represent the transpose and inverse of a square matrix, respectively. Define $\mathbb{R}^d$ as the real space in dimension of $d > 0$. Let $0_{d\times d}$ be the $d \times d$ zero matrix and $I_{d\times d}$ be the identity matrix, for example, $I_{2\times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Define $\otimes$ as the Kronecker product and $\|\cdot\|$ as the Euclidean vector norm. Let $P > 0$ (resp. $P < 0$) denote that P is symmetric and positive definite (resp. negative definite). Define $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ respectively as the largest and smallest eigenvalue of a matrix. For MASs with W agents, we define $x_i \in \mathbb{R}^d$ as the position of agent i. Define the position of all agents $x = \left[x_1^{\mathrm{T}}, \ \dots \ , x_W^{\mathrm{T}}\right]^{\mathrm{T}}$ as a column vector in $\mathbb{R}^{Wd}$.

2.2 | Basic graph theory

A formation in $\mathbb{R}^d$, denoted as $(\mathcal{G}, x)$, is the graph $\mathcal{G}$ embedded with a configuration x in $\mathbb{R}^{Wd}$ where its vertex i is corresponding to point x_i. An undirected graph $\mathcal{G} = (\mathcal{V}, \varepsilon)$ is used to describe the interaction among agents, which consists of a vertex set $\mathcal{V} = \{1, 2, \ \dots \ , W\}$ and an edge set $\varepsilon \subseteq \mathcal{V} \times \mathcal{V}$. The set of neighbors of vertex i is denoted by $\mathcal{N}_i = \{j \in \mathcal{V} : (i,j) \in \varepsilon\}$.

Denote the constant configuration $r = \left[r_1^{\mathrm{T}}, \ \dots \ , r_W^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{R}^{Wd}$ as a nominal configuration, which is a basic geometric pattern that all the agents want to maintain. Define the nominal formation as $(\mathcal{G}, r)$. Then, the target formation $x^* \in \mathbb{R}^{Wd}$ with maneuvers can be described by

$$x^*(t) = [I_{W\times W} \otimes E(t)]r + 1_W \otimes b(t), \tag{1}$$

where $1_W \in \mathbb{R}^W$ is the column vector with all entries equal to one, $b(t) \in \mathbb{R}^d$ represents the formation's translational maneuver, and $E(t) \in \mathbb{R}^{d\times d}$ can be used to realize the whole formation's rotational, scaling, and shearing maneuvers with respect to the nominal configuration r. Thus, the desired position $x_i^* \in \mathbb{R}^d$ of agent i in the target formation is

$$x_i^*(t) = E(t)r_i + b(t), \tag{2}$$

where r_i is the position of agent i in the nominal configuration r.

2.3 | Stress matrix

Consider the MASs consisting of N leaders and M followers. We denote the set of leaders as $\mathcal{V}_l$ and the set of followers as $\mathcal{V}_f$. Define the positions of leaders $x_l = \left[x_1^{\mathrm{T}}, \ \dots \ , x_N^{\mathrm{T}}\right]^{\mathrm{T}}$ as a column vector in $\mathbb{R}^{Nd}$; Define the positions of followers $x_f = \left[x_{N+1}^{\mathrm{T}}, \ \dots \ , x_{N+M}^{\mathrm{T}}\right]^{\mathrm{T}}$ as a column vector in $\mathbb{R}^{Md}$.

A stress of the formation $(\mathcal{G}, x)$ represents a set of scalars $\left\{\omega_{ij}\right\}_{(i,j)\in\varepsilon}$ with $\omega_{ij} = \omega_{ji} \in \mathbb{R}$, which is assigned to all edges. We define the stress matrix as $\Omega \in \mathbb{R}^{(M+N)\times(M+N)}$, which satisfies

$$\left[\Omega\right]_{ij} = \begin{cases} 0, & i \neq j, (i,j) \notin \varepsilon; \\ -\omega_{ij}, & i \neq j, (i,j) \in \varepsilon; \\ \Sigma_{k\in\mathcal{N}_i}\omega_{ik}, & i = j; \end{cases} \tag{3}$$

where the edge $(i,j) \in \varepsilon$ indicates that agent i can communicate information with agent j.

A stress is called an equilibrium if it satisfies $\sum_{j\in\mathcal{N}_i} \omega_{ij}\left(x_i - x_j\right) = 0$, which can be written in a compact form as $(\Omega \otimes I_{d\times d})x = 0$.

According to the partition of leaders and followers, the stress matrix Ω can be partitioned as

$$\Omega = \begin{bmatrix} \Omega_{ll} & \Omega_{lf} \\ \Omega_{fl} & \Omega_{ff} \end{bmatrix}, \tag{4}$$

where $\Omega_{ff} \in \mathbb{R}^{M\times M}$, $\Omega_{fl} \in \mathbb{R}^{M\times N}$, $\Omega_{lf} \in \mathbb{R}^{N\times M}$ and $\Omega_{ll} \in \mathbb{R}^{N\times N}$.

The affine image of the nominal configuration is defined as

$$\mathcal{A}(r) = \left\{x \in \mathbb{R}^{d(M+N)} : x = \left(I_{(M+N)\times(M+N)} \otimes E\right) r + 1_{(M+N)} \otimes b\right\}. \tag{5}$$

Therefore, it is clear that the target formation x^* is the affine transformation of the nominal configuration r.

Definition 1 (Affine Span). Given a set of points $\{x_i\}_{i=1}^{M+N}$ in $\mathbb{R}^d$, the affine span $\mathcal{W}$ of these points is

$$\mathcal{W} = \left\{\sum_{i=1}^{M+N} a_i x_i : a_i \in \mathbb{R} \text{ for all } i \text{ and } \sum_{i=1}^{M+N} a_i = 1\right\}.$$

Definition 2 (Affine Localizability). The nominal formation $(\mathcal{G}, r)$ is affinely localizable by the leaders if for any $x = \left[x_l^{\mathrm{T}}, x_f^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathcal{A}(r)$, x_f can be uniquely determined by x_l.

As used in Reference 16, we introduce an assumption and a lemma.

Assumption 1. We assume that

(1) the nominal configuration r affinely span $\mathbb{R}^d$.
(2) the nominal formation $(\mathcal{G}, r)$ has a positive semidefinite stress matrix Ω satisfying $\mathrm{rank}(\Omega) = M + N - 1$.
(3) the nominal formation $(\mathcal{G}, r)$ is affinely localizable by the leaders.

More details about Assumption 1 can be found in References 16 and 29. To intuitively explain the graphic conditions provided in Assumption 1, a sample example is given in Figure 1. There are five vertices in Figure 1A,B, which both satisfy Assumption 1(1). Note that Figure 1B is universally rigid which is the necessary and sufficient condition of Assumption 1(2), while Figure 1A is not because the positions of followers 4, 5 in Figure 1A are not uniquely determined in $\mathbb{R}^3$. Therefore, Figure 1A does not satisfy Assumption 1(2)–(3) but Figure 1B does.

Lemma 1 (16, Theorem 2). *If Assumption 1 holds, then Ω_{ff} is positive definite, and for any $x = \left[x_l^T, x_f^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathcal{A}(r)$, x_f can be uniquely calculated as*

$$x_f = -\left(\Omega_{ff}^{-1}\Omega_{fl} \otimes I_{d\times d}\right) x_l. \tag{6}$$

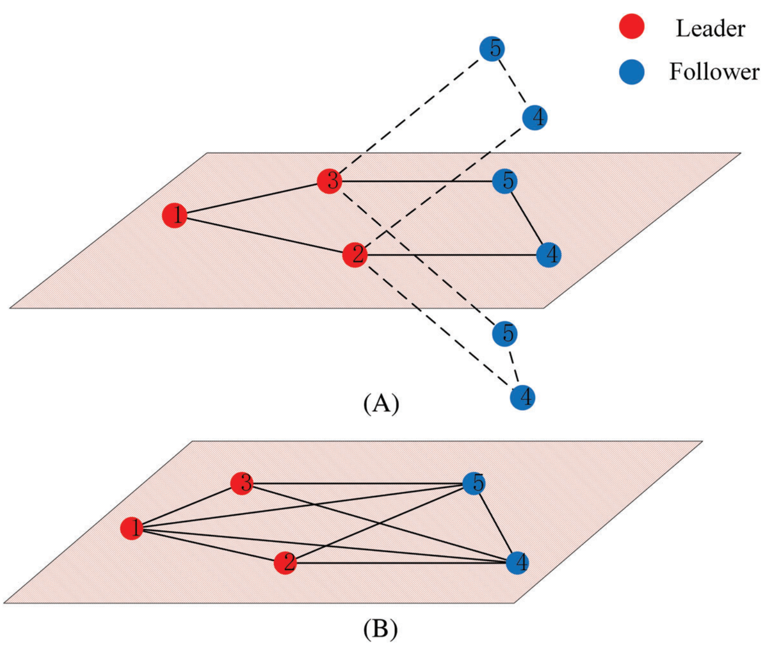

FIGURE 1 (A) A framework that does not satisfy the Assumption 1. (B) A framework that satisfies the Assumption 1

2.4 Optimality

The design of the optimal control law in this article is based on the following two lemmas. Consider a nonlinear dynamical system in form of

$$\dot{x} = f(x) + g(x)u, \tag{7}$$

where $f(\cdot) : \mathbb{R}^d \mapsto \mathbb{R}^d$, $g(\cdot) : \mathbb{R}^d \mapsto \mathbb{R}^d \times \mathbb{R}^p$ and $u \in \mathbb{R}^p$.

Lemma 2 (Inverse optimality[30]). *Consider the dynamical system (7). Let $u = \phi(x)$ be a stabilizing control law with respect to a manifold $\mathcal{S}$. Given the infinite horizon optimality criterion*

$$J(x_0, u) = \int_0^\infty \mathcal{L}(x, u)\mathrm{d}t. \tag{8}$$

If there exist scalar functions $V(x)$, $L_1(x)$, and $\mathcal{K}$ class functions α, γ satisfying the following conditions:

$$\begin{aligned} V(x) &= 0 \Leftrightarrow x \in \mathcal{S} \\ V(x) &\geq \alpha(d(x, \mathcal{S})) \\ L_1(x) &\geq \gamma(d(x, \mathcal{S})) \\ L_1(x) + \nabla V(x)^{\mathrm{T}} f(x) &= \frac{1}{4}[\nabla V(x)]^{\mathrm{T}} g(x) R^{-1} g^{\mathrm{T}}(x) \nabla V(x) \\ \phi(x) &= -\frac{1}{2} R^{-1} g(x)^{\mathrm{T}} \nabla V(x), \end{aligned} \tag{9}$$

where $d(x, \mathcal{S})$ is the distance from the point x to the manifold $\mathcal{S}$, then $u = \phi(x)$ is optimal with respect to the performance index J with the integrand $\mathcal{L} = L_1(x) + u^{\mathrm{T}} R u$. Moreover, the optimal value of the performance criterion equals $J(x_0, \phi(x)) = V(x_0)$.

Lemma 3 (Optimality[31]). *Consider the dynamical system (7). The control vector u can minimize the following performance index:*

$$J(x, u) = \int_0^\infty e^{rt} \mathcal{L}(x, u)\mathrm{d}t, \tag{10}$$

where $r > 0$ is the stability degree, if there exists a Hamilton multiplier $m(t)$ satisfying the following conditions:

$$\frac{\partial H}{\partial u} = 0, \tag{11}$$

$$\frac{\partial H}{\partial x} = -rm - \dot{m}, \tag{12}$$

where the Hamilton equation $H(x, u) = \mathcal{L}(x, u) + m(t) f(x, u)$.

Remark 1. In the linear quadratic case, the Hamilton–Jacobi–Bellman (HJB) equation (9) becomes the algebraic Riccati equation (ARE).

3 INVERSE OPTIMAL AFFINE CONTROL ALGORITHM DESIGN

In this article, we assume that the dynamics of leaders are given by

$$\dot{x}_i = A(x_i - x_i^*), \quad i \in \mathcal{V}_l, \tag{13}$$

where $A \in \mathbb{R}^{d\times d}$ and the target position $x_i^* \in \mathbb{R}^d$ of leader $i$is a constant vector. All followers are governed by the following linear dynamics

$$\dot{x}_i = Ax_i + Bu_i, \quad i \in \mathcal{V}_f, \tag{14}$$

where $u_i \in \mathbb{R}^p$ is the controller input of the follower i, $B \in \mathbb{R}^{d\times p}$ and the pair (A, B) is controllable.

Next, we formulate the problem to be solved.

Problem 1. Design a cooperative protocol u_i of system (14) such that some performance indices can be optimized and simultaneously each follower tracks its target position in the affine formation, that is,

$$\lim_{t\to\infty}(x_i(t) - x_i^*) = 0,\ i \in \mathcal{V}_f, \tag{15}$$

where x_i^* denotes the desired position of agent i in the target formation.

To solve this problem, we first define the local neighborhood tracking error as

$$\delta_i = \sum_{j=1}^{M+N} \omega_{ij}(x_i - x_j),\ i \in \mathcal{V}_f. \tag{16}$$

Writing the local neighborhood tracking error into a compact form yields

$$\delta_f = \left(\Omega_{ff} \otimes I_{d\times d}\right) x_f + \left(\Omega_{fl} \otimes I_{d\times d}\right) x_l, \tag{17}$$

where $\delta_f = \left[\delta_{1+N}^{\mathrm{T}}, \dots, \delta_{M+N}^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{R}^{Md}$.

To design the optimal proportional integral control law, we give the augmented system of the local neighborhood tracking error as following.

$$\begin{aligned}\dot{\xi}_f &= I_{M\times M} \otimes \begin{bmatrix} 0_{d\times d} & I_{d\times d} \\ 0_{d\times d} & A \end{bmatrix} \xi_f + \Omega_{ff} \otimes \begin{bmatrix} 0_{d\times p} \\ B \end{bmatrix} \dot{u}_f \\ &= I_{M\times M} \otimes \overline{A}\xi_f + \Omega_{ff} \otimes \overline{B}\dot{u}_f, \end{aligned}\tag{18}$$

where $u_f = \left[u_{1+N}^{\mathrm{T}}, \dots, u_{M+N}^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{R}^{Mp}$, $\xi_f = \left[\xi_{1+N}^{\mathrm{T}}, \dots, \xi_{M+N}^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{R}^{2Md}$, $\xi_i = \left[\delta_i^{\mathrm{T}}, \dot{\delta}_i^{\mathrm{T}}\right]^{\mathrm{T}}, i = N+1, \dots, N+M$.

We propose an optimal affine formation control law in the form of

$$u_f(t) = -G\int_0^t \xi_f(\tau)\mathrm{d}\tau, \tag{19}$$

where $G \in \mathbb{R}^{pM\times 2dM}$ is the feedback gain matrix. Note that u_f is a proportional-integral (PI) control law.

To guarantee followers can track their target positions, the error dynamical system (18) should asymptotically converge to the origin.

Now, we present the main result which gives sufficient conditions for the global optimality of the linear control law (19).

Theorem 1. *For error dynamics (18), suppose that Assumption 1 holds and there exist positive define matrices $P_1 \in \mathbb{R}^{M\times M}$ and $P_2 \in \mathbb{R}^{2p\times 2p}$ satisfying*

$$P_1 = c(R_1\Omega_{ff}^{-1} + \Omega_{ff}^{-1}R_1), \tag{20}$$

$$\overline{A}^{\mathrm{T}}P_2 + P_2\overline{A} + Q_2 - P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2 = 0, \tag{21}$$

for some symmetric positive definite matrices $Q_2 \in \mathbb{R}^{2d\times 2d}, R_1 \in \mathbb{R}^{M\times M}, R_2 \in \mathbb{R}^{p\times p}$, and constant coupling gain $c > 0$. If G and c further satisfy the following conditions:

$$G = c(I_{M\times M} + R_1^{-1}\Omega_{ff}R_1\Omega_{ff}^{-1}) \otimes R_2^{-1}\overline{B}^{\mathrm{T}}P_2, \tag{22}$$

$$\begin{aligned}&\lambda_{\min}(c(R_1 + \Omega_{ff}R_1\Omega_{ff}^{-1} + \Omega_{ff}^{-1}R_1\Omega_{ff} + \Omega_{ff}^{-1}R_1\Omega_{ff}R_1^{-1}\Omega_{ff}R_1\Omega_{ff}^{-1}) \otimes P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2 \\ &\quad + (R_1\Omega_{ff}^{-1} + \Omega_{ff}^{-1}R_1) \otimes (Q_2 - P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2)) > 0,\end{aligned}\tag{23}$$

then the control $u_f(t) = -G\int_0^t \xi_f(\tau)\mathrm{d}\tau$ *achieves the objective (15) and is globally optimal with respect to the performance index:*

$$J = \int_0^\infty \{\xi_f^{\mathrm{T}}[c^2(R_1 + \Omega_{ff}R_1\Omega_{ff}^{-1} + \Omega_{ff}^{-1}R_1\Omega_{ff} + \Omega_{ff}^{-1}R_1\Omega_{ff}R_1^{-1}\Omega_{ff}R_1\Omega_{ff}^{-1}) \otimes P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2 \\ + c(R_1\Omega_{ff}^{-1} + \Omega_{ff}^{-1}R_1) \otimes (Q_2 - P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2)]\xi_f + \dot{u}_f^{\mathrm{T}}(R_1 \otimes R_2)\dot{u}_f\}\mathrm{d}t. \tag{24}$$

Proof. Define the matrix $P = P_1 \otimes P_2$. Since P_1 and P_2 are nonsingular, P is also nonsingular. Define the Lyapunov function as

$$V(\xi_f) = \xi_f^{\mathrm{T}}P\xi_f > 0. \tag{25}$$

It can be verified that (25) satisfies the conditions in Lemma 2. According to (24), we have

$$\begin{aligned} L_1(\xi_f) &= \xi_f^{\mathrm{T}}Q\xi_f \\ &= \xi_f^{\mathrm{T}}\left[c^2\left(R_1 + \Omega_{ff}R_1\Omega_{ff}^{-1} + \Omega_{ff}^{-1}R_1\Omega_{ff} + \Omega_{ff}^{-1}R_1\Omega_{ff}R_1^{-1}\Omega_{ff}R_1\Omega_{ff}^{-1}\right) \otimes P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2 \right. \\ &\quad \left. + c\left(R_1\Omega_{ff}^{-1} + \Omega_{ff}^{-1}R_1) \otimes (Q_2 - P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2\right)\right]\xi_f. \end{aligned} \tag{26}$$

By using (23), one has

$$Q > 0. \tag{27}$$

According to (9), the ARE for the dynamical system (18) is

$$\left(I_{M\times M} \otimes \overline{A}\right)^{\mathrm{T}}P + P\left(I_{M\times M} \otimes \overline{A}\right) + Q - P(\Omega_{ff} \otimes \overline{B})R^{-1}\left(\Omega_{ff} \otimes \overline{B}\right)^{\mathrm{T}}P = 0, \tag{28}$$

where $R = R_1 \otimes R_2 \in \mathbb{R}^{pM\times pM}$.

Then (28) can also be written as

$$P_1 \otimes \overline{A}^{\mathrm{T}}P_2 + P_1 \otimes P_2\overline{A} + Q - \left(P_1\Omega_{ff} \otimes P_2\overline{B}\right)\left(R_1^{-1} \otimes R_2^{-1}\right)\left(\Omega_{ff}P_1 \otimes \overline{B}^{\mathrm{T}}P_2\right) = 0 \tag{29}$$

which implies that

$$P_1 \otimes \left(\overline{A}^{\mathrm{T}}P_2 + P_2\overline{A}\right) + Q - \left(P_1\Omega_{ff}R_1^{-1}\Omega_{ff}P_1\right) \otimes \left(P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2\right) = 0. \tag{30}$$

Substituting (20) in (30) results in

$$\begin{aligned} Q = c^2&\left(R_1 + \Omega_{ff}R_1\Omega_{ff}^{-1} + \Omega_{ff}^{-1}R_1\Omega_{ff} + \Omega_{ff}^{-1}R_1\Omega_{ff}R_1^{-1}\Omega_{ff}R_1\Omega_{ff}^{-1}\right) \otimes P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2 \\ &+ c(R_1\Omega_{ff}^{-1} + \Omega_{ff}^{-1}R_1) \otimes (Q_2 - P_2\overline{B}R_2^{-1}\overline{B}^{\mathrm{T}}P_2). \end{aligned} \tag{31}$$

By using conditions (20) and (28), the optimal control input of system (18) can be written by

$$\begin{aligned} \dot{u}_f &= -R^{-1}(\Omega_{ff} \otimes \overline{B})^{\mathrm{T}}P\xi_f \\ &= -\left(R_1^{-1} \otimes R_2^{-1}\right)\left(\Omega_{ff} \otimes \overline{B}\right)^{\mathrm{T}}(P_1 \otimes P_2)\,\xi_f \\ &= -R_1^{-1}\Omega_{ff}P_1 \otimes R_2^{-1}\overline{B}^{\mathrm{T}}P_2\xi_f. \\ &= -c\left(I_{M\times M} + R_1^{-1}\Omega_{ff}R_1\Omega_{ff}^{-1}\right) \otimes R_2^{-1}\overline{B}^{\mathrm{T}}P_2\xi_f. \end{aligned} \tag{32}$$

Based on (22), we can get that the control law (19) is globally optimal with respect to the performance index (24).

By combining (20), (22), and (30), taking the time-derivative of the Lyapunov function (25) gives

$$
\begin{aligned}
\dot{V}(\xi_f) &= 2\xi_f^{\mathrm{T}} P \dot{\xi}_f \\
&= 2\xi_f^{\mathrm{T}} P (I_{M\times M} \otimes \overline{A} - c \left(\Omega_{ff} + \Omega_{ff} R_1^{-1} \Omega_{ff} R_1 \Omega_{ff}^{-1}\right) \otimes \overline{B} R_2^{-1} \overline{B}^{\mathrm{T}} P_2) \xi_f \\
&= \xi_f^{\mathrm{T}} \left(P_1 \otimes \left(P_2 \overline{A} + \overline{A}^{\mathrm{T}} P_2\right) - 2 \left(P_1 \Omega_{ff} R_1^{-1} \Omega_{ff} P_1\right) \otimes \left(P_2 \overline{B} R_2^{-1} \overline{B}^{\mathrm{T}} P_2\right) \xi_f \right. \\
&= -\xi_f^{\mathrm{T}} \left(Q + P \left(\Omega_{ff} \otimes \overline{B}\right) \left(R_1^{-1} \otimes R_2^{-1}\right) \left(\Omega_{ff} \otimes \overline{B}\right)^{\mathrm{T}} P\right) \xi_f < 0
\end{aligned} \tag{33}
$$

which implies that $\lim_{t\to\infty} \xi_f(t) = 0$. The conditions in Lemma 2 are all satisfied. This completes the proof. ■

Remark 2. Note that for a large-scale network, the dimension of the corresponding matrices R_1 and P_1 becomes very large. To calculate the eigenvalues of these large matrices, the method of minimized iterations[32] and the generalization of the Davidson's method[33] can be employed.

Remark 3. The control law (19) does not explicitly contain global information Ω_{ff}. However, according to (23) the control parameter c in (22) is related with the global information Ω_{ff}. Although formation control design stage is centralized, the implementation stage is fully distributed.

Remark 4. Different from Reference 34, the dynamics of the leaders are considered as the form of (13). Moreover, to optimally stabilize the dynamics of tracking error system (18), we proposed a PI control law (19), which yields the different performance index (24) as compared to Reference 34. In addition, the limitation of inverse optimal control is that the performance indicators cannot be flexibly given, but determined by the proposed control law. The weight of performance index needs to be adjusted according to the matrices Q_2, R_1, and R_2. To have more flexible performance index, the optimal control law will be designed in the next section.

4 | OPTIMAL AFFINE CONTROL ALGORITHM DESIGN

In Section 3, the optimal performance index (24) cannot be freely selected which is determined by the designed control law (19), thus is called inverse optimality. So in this section, we propose a new optimal affine algorithm to minimize a prespecified performance index.

Next, we formulate the problem to be solved in this section.

Problem 2. Design the optimal control law u_i of system (14) such that the followers converge to their desired target positions and the following performance index is optimal.

$$
J_i = \frac{1}{2} \int_0^\infty e^{\gamma_i t} \left(e_i^{\mathrm{T}} Q_i e_i + u_i^{\mathrm{T}} R_i u_i\right) \mathrm{d}t, i \in \mathcal{V}_f, \tag{34}
$$

where $e_i = x_i - x_i^*$, Q_i, and R_i are positive definite matrices and $\gamma_i > 0$ is the stability degree.

4.1 | The design of cooperative estimator with proportional-integral feedback

First, we need to design an estimator to estimate the target positions of followers. We denote $\hat{x}_i \in \mathbb{R}^d$ as the estimation of the position x_i. For the design of cooperative estimators, we can similarly define the neighborhood estimation error for agent i as

$$
\zeta_i = -\sum_{j=1}^{M+N} \omega_{ij} \left(\hat{x}_i - \hat{x}_j\right), \ i \in \mathcal{V}_f. \tag{35}
$$

Note that $\hat{x}_j = x_j, j \in \mathcal{V}_l$. The neighborhood estimation error can also be written in a compact form

$$\zeta_f = -(\Omega_{ff} \otimes I_{d\times d})\hat{x}_f - (\Omega_{fl} \otimes I_{d\times d})\hat{x}_l \tag{36}$$

with $\hat{x}_l = \left[\hat{x}_1^{\mathrm{T}}, \ldots, \hat{x}_N^{\mathrm{T}}\right]^{\mathrm{T}}$ and $\hat{x}_f = \left[\hat{x}_{N+1}^{\mathrm{T}}, \ldots, \hat{x}_{N+M}^{\mathrm{T}}\right]^{\mathrm{T}}$.

The cooperative estimator for each follower $i(i \in \mathcal{V}_f)$ is designed in the form of

$$\dot{\hat{x}}_i = A\hat{x}_i + F\left(\zeta_i + v_i\right), \tag{37}$$

where $F \in \mathbb{R}^{d\times d}$ is the estimator gain and $\dot{v}_i = \zeta_i$. Then the cooperative estimator dynamics can be written in a compact form

$$\dot{\hat{x}}_f = (I_{M\times M} \otimes A)\,\hat{x}_f + (I_{M\times M} \otimes F)\,\zeta_f + (I_{M\times M} \otimes F)\,v_f \tag{38}$$

with $\zeta_f = \left[\zeta_{N+1}^{\mathrm{T}}, \ldots, \zeta_{N+M}^{\mathrm{T}}\right]^{\mathrm{T}}$ and $v_f = \left[v_{N+1}^{\mathrm{T}}, \ldots, v_{N+M}^{\mathrm{T}}\right]^{\mathrm{T}}$.

Denote the state estimation error as

$$\overline{x}_f = \hat{x}_f - x_f^*. \tag{39}$$

Then based on Lemma 1 the state estimation error can be described as

$$\overline{x}_f = \hat{x}_f + (\Omega_{ff}^{-1}\Omega_{fl} \otimes I_{d\times d})x_l. \tag{40}$$

Based on the conditions (13) and (38), the state estimation error dynamics can be written as

$$\begin{aligned}\dot{\overline{x}}_f &= \dot{\hat{x}}_f + \left(\Omega_{ff}^{-1}\Omega_{fl} \otimes I_{d\times d}\right)\dot{x}_l \\ &= A_e\overline{x}_f + (I_{M\times M} \otimes F)\,v_f - \left(\Omega_{ff}^{-1}\Omega_{fl} \otimes A\right)x_l^*\end{aligned} \tag{41}$$

with $A_e = I_{M\times M} \otimes A - \left(\Omega_{ff} \otimes F\right)$ and $\dot{v}_f = \zeta_f$.

Lemma 4. *Under the Assumption 1, the state estimation error $\overline{x}_f$ under the estimator (37) converges to zero globally and exponentially if there exists a matrix F such that*

$$\Omega_{ff} \otimes F - I_{M\times M} \otimes A + \left(\Omega_{ff} \otimes F - I_{M\times M} \otimes A\right)^{\mathrm{T}} > 0 \tag{42}$$

and

$$\Omega_{ff} \otimes F + \left(\Omega_{ff} \otimes F\right)^{\mathrm{T}} > 0. \tag{43}$$

Proof. According to (41), we can have the augmented estimation error dynamics:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\overline{x}_f \\ \dot{\overline{x}}_f\end{bmatrix} &= \begin{bmatrix}0_{dM\times dM} & I_{dM\times dM} \\ -\Omega_{ff} \otimes F & A_e\end{bmatrix}\begin{bmatrix}\overline{x}_f \\ \dot{\overline{x}}_f\end{bmatrix} + \begin{bmatrix}0_{Md\times Nd} \\ -\left(\Omega_{ff}^{-1}\Omega_{fl} \otimes A\right)\end{bmatrix}\dot{x}_l^* \\ &= \overline{A}_e\begin{bmatrix}\overline{x}_f \\ \dot{\overline{x}}_f\end{bmatrix} + L_e\dot{x}_l^*.\end{aligned} \tag{44}$$

Since x_l^* is a constant vector, that is, $\dot{x}_l^* = 0$, the state estimation error can converge to the origin if $\overline{A}_e$ is Hurwitz. Define the eigenvalue of $\overline{A}_e$ as $\lambda \in \mathbb{C}$. Define the eigenvector of $\overline{A}_e$ as $w \in \mathbb{R}^{2dM}$. Then one has

$$w^{\mathrm{T}}\left(\overline{A}_e - \lambda I_{2dM\times 2dM}\right)w = 0 \tag{45}$$

which implies that

$$\lambda^2 w^{\mathrm{T}} I_{dM\times dM} w + \lambda w^{\mathrm{T}} \left(\Omega_{ff} \otimes F - I_{M\times M} \otimes A\right) w + w^{\mathrm{T}} \left(\Omega_{ff} \otimes F\right) w = 0. \tag{46}$$

It follows that λ only has negative real parts if $w^{\mathrm{T}}(\Omega_{ff} \otimes F - I_{M\times M} \otimes A)w$ and $w^{\mathrm{T}}(\Omega_{ff} \otimes F)w$ are both positive. Note that

$$w^{\mathrm{T}}\left(\Omega_{ff} \otimes F - I_{M\times M} \otimes A + (\Omega_{ff} \otimes F - I_{M\times M} \otimes A)^{\mathrm{T}}\right)w = 2w^{\mathrm{T}} \left(\Omega_{ff} \otimes F - I_{M\times M} \otimes A\right) w \tag{47}$$

and

$$w^{\mathrm{T}} \left(\Omega_{ff} \otimes F + \left(\Omega_{ff} \otimes F\right)^{\mathrm{T}}\right) w = 2w^{\mathrm{T}} \left(\Omega_{ff} \otimes F\right) w. \tag{48}$$

So λ only has negative real parts if $\Omega_{ff} \otimes F - I_{M\times M} \otimes A + (\Omega_{ff} \otimes F - I_{M\times M} \otimes A)^{\mathrm{T}}$ and $\Omega_{ff} \otimes F + (\Omega_{ff} \otimes F)^{\mathrm{T}}$ are both positive definite. Note that the conditions (42) and (43) actually are linear matrix inequalities, which can be checked by numerical tools(e.g., feasp, mincx, and gevp) in the system design stage so that the solution F can be obtained. This completes the proof. ■

Remark 5. Although Lemma 4 does not give the condition about the estimate gain F directly, it is easy to select the F such that (42)–(43) hold. The matrices of (42) and (43) are already symmetric matrices and Ω_{ff} is the positive definite matrix. Thus, choosing a big positive definite matrix as F usually makes (42)–(43) hold.

4.2 | The design of optimal formation control law

To design the optimal control law, we need to modify the performance index (34) firstly. If $\hat{x}_i - x_i^*$ can converge to 0, that is, $\zeta_i \to 0$, the tracking error $e_i = x_i - x_i^* \to x_i - \hat{x}_i$. Then (34) can be approximated as the following performance index which tracks the estimation of the position $\hat{x}_i$.

$$J_i = \int_0^{\infty} e^{\gamma_i t} r(\tilde{x}_i, u_i)\mathrm{d}t \tag{49}$$

with the utility function

$$r(\tilde{x}_i, u_i) = \frac{1}{2}\left(\tilde{x}_i^{\mathrm{T}} Q_i \tilde{x}_i + u_i^{\mathrm{T}} R_i u_i\right), \tag{50}$$

where $\tilde{x}_i = x_i - \hat{x}_i$.

To guarantee that $\tilde{x}_i = 0$, we modify the performance index (49) as the following:[35]

$$J_i = \frac{1}{2}\int_0^{\infty} e^{\gamma_i t} \left(\tilde{x}_i^{\mathrm{T}} Q_i \tilde{x}_i + \dot{u}_i^{\mathrm{T}} R_i \dot{u}_i\right) \mathrm{d}t. \tag{51}$$

Under the estimator (37), it follows that

$$\begin{aligned}\dot{\tilde{x}}_i &= A\tilde{x}_i + Bu_i - F\zeta_i - Fv_i,\\ \ddot{\tilde{x}}_i &= A\dot{\tilde{x}}_i + Bd_i - F\dot{\zeta}_i - F\zeta_i,\end{aligned} \tag{52}$$

where $d_i = \dot{u}_i$.

For simplicity, (52) and (53) can be written in a compact form as:

$$\begin{aligned}\dot{\eta}_i &= \begin{bmatrix} 0_{d\times d} & I_{d\times d} \\ 0_{d\times d} & A \end{bmatrix} \eta_i + \begin{bmatrix} 0_{d\times p} \\ B \end{bmatrix} d_i + \begin{bmatrix} 0_{d\times d} & 0_{d\times d} \\ -F & -F \end{bmatrix} \begin{bmatrix} \zeta_i \\ \dot{\zeta}_i \end{bmatrix}\\ &= \overline{A}\eta_i + \overline{B}d_i + \overline{F\zeta}_i,\end{aligned} \tag{54}$$

where $\eta_i = \begin{bmatrix} \tilde{x}_i^{\mathrm{T}} & \dot{\tilde{x}}_i^{\mathrm{T}} \end{bmatrix}^{\mathrm{T}}, \overline{\zeta}_i = \begin{bmatrix} \zeta_i^{\mathrm{T}} & \dot{\zeta}_i^{\mathrm{T}} \end{bmatrix}^{\mathrm{T}}, i \in \mathcal{V}_f$.

Then, the performance index (51) can be written as

$$J_i = \frac{1}{2}\int_0^{\infty} e^{\gamma_i t}\left(\eta_i^{\mathrm{T}}\overline{Q}_i\eta_i + d_i^{\mathrm{T}}R_i d_i\right)\mathrm{d}t \tag{55}$$

with

$$\overline{Q}_i = \begin{bmatrix} Q_i & 0_{d\times d} \\ 0_{d\times d} & 0_{d\times d}\end{bmatrix}. \tag{56}$$

Next, we show the main result which gives the optimal control protocol with respect to the above performance index (55).

Theorem 2. *Under the estimator (37), the neighborhood estimation error $\overline{\zeta}_i$ can converge to zero vector, let u_i take the form of*

$$u_i = -R_i^{-1}\overline{B}^{\mathrm{T}}P_i^*[z_i^{\mathrm{T}} \quad \tilde{x}_i^{\mathrm{T}}]^{\mathrm{T}}, i \in \mathcal{V}_f, \tag{57}$$

where $\dot{z}_i = \tilde{x}_i$ and P_i^* satisfies the ARE

$$0 = \overline{A}^{\mathrm{T}}P_i^* + P_i^*\overline{A} + \gamma_i P_i^* - P_i^*\overline{B}R_i^{-1}\overline{B}^{\mathrm{T}}P_i^* + \overline{Q}_i. \tag{58}$$

Then the admissible control u_i can minimize the performance index (55).

Proof. According to the Lemma 3, we define the Hamiltonian function as

$$H = 0.5\left(\eta_i^{\mathrm{T}}\overline{Q}_i\eta_i + d_i^{\mathrm{T}}R_i d_i) + m_i^{\mathrm{T}}(t)(\overline{A}\eta_i + \overline{B}d_i + \overline{F\zeta}_i\right), \tag{59}$$

where $m_i(t) \in \mathbb{R}^d$ is the Hamiltonian multiplier.

Employing the stationary condition (11), we can get that

$$\frac{\partial H}{\partial d_i} = \overline{B}^{\mathrm{T}}m_i(t) + R_i d_i = 0. \tag{60}$$

We assume that $m_i(t) = P_i^*\eta_i(t), P_i^* \in \mathbb{R}^{d\times d}$. Using the condition (12), we can have

$$\frac{\partial H}{\partial \eta_i} = \overline{Q}_i\eta_i + \overline{A}^{\mathrm{T}}m_i = -\gamma_i m_i - \dot{m}_i. \tag{61}$$

Differentiating m_i and performing some manipulations to (61) yield

$$\dot{m}_i = P_i^*\dot{\eta}_i = -\gamma_i m_i - \overline{Q}_i\eta_i - \overline{A}^{\mathrm{T}}m_i. \tag{62}$$

Substituting (54) into (62) results in

$$0 = P_i^*\overline{F\zeta}_i + \left(\overline{A}^{\mathrm{T}}P_i^* + P_i^*\overline{A} + \gamma_i P_i^* - P_i^*\overline{B}R_i^{-1}\overline{B}^{\mathrm{T}}P_i^* + \overline{Q}_i\right)\eta_i. \tag{63}$$

Therefore, the above holds for $\forall\eta_i$ if and only if (58) is satisfied with the condition that $\overline{\zeta}_i = 0$. At the same time, based on (60) the time derivative of control input u_i is given as

$$d_i = -R_i^{-1}\overline{B}^{\mathrm{T}}P_i^*\eta_i = -R_i^{-1}\overline{B}^{\mathrm{T}}P_i^*\begin{bmatrix}\tilde{x}_i \\ \dot{\tilde{x}}_i\end{bmatrix}. \tag{64}$$

Then the control input u_i is given as

$$u_i(t) = -R_i^{-1}\overline{B}^{\mathrm{T}} P_i^* \begin{bmatrix} \int_0^t \tilde{x}_i(\tau)\mathrm{d}\tau \\ \tilde{x}_i(t) \end{bmatrix}. \tag{65}$$

This completes the proof. ■

Theorem 3. *Under the assumption that $\gamma_i > 0$, suppose the optimal control law (57) is applied to the augmented system (54). Then, one has:*

1) *The homogeneous ARE (58) has a unique positive semidefinite solution.*
2) *The error $\tilde{x}_i = x_i - \hat{x}_i$ goes to zero asymptotically.*

Proof. 1) Note that the ARE (58) can be written as

$$0 = \left(\overline{A} + 0.5\gamma_i I_{2d\times 2d}\right)^{\mathrm{T}} P_i^* + P_i^* \left(\overline{A} + 0.5\gamma_i I_{2d\times 2d}\right) - P_i^* \overline{B} R_i^{-1} \overline{B}^{\mathrm{T}} P_i^* + \overline{Q}_i. \tag{66}$$

This is the same as the ARE for the general linear system dynamics given by $\overline{A} + 0.5\gamma_i I_{2d\times 2d}$ and $\overline{B}$. Therefore, a unique solution to the ARE (66) exists if $(\overline{A} + 0.5\gamma_i I_{2d\times 2d}, \overline{B})$ is controllable. This requires that $(A + 0.5\gamma_i I_{d\times d}, B)$ is controllable. However, since (A, B) is controllable, then $(A + 0.5\gamma_i I_{d\times d}, B)$ is also controllable for any $\gamma_i > 0$. This completes the proof.

2) The closed-loop augmented error system is given as

$$\begin{aligned} \dot{\eta}_i &= \left(\overline{A} - \overline{B} R_i^{-1} \overline{B}^{\mathrm{T}} P_i^*\right) \eta_i + \overline{F\zeta}_i \\ &= A_{ci}\eta_i + \overline{F\zeta}_i. \end{aligned} \tag{67}$$

where $A_{ci} = \overline{A} - \overline{B} R_i^{-1} \overline{B}^{\mathrm{T}} P_i^*$. Note that the ARE (58) can be written as

$$A_{ci}^{\mathrm{T}} P_i^* + P_i^* A_{ci} = -\overline{Q}_i - \gamma_i P_i^* - P_i^* \overline{B} R_i^{-1} \overline{B}^{\mathrm{T}} P_i^* < 0. \tag{68}$$

Therefore, the system (67) is asymptotically stable. According to Lemma 4, we can have $\overline{\zeta}_i \to 0$. So it follows that $\tilde{x}_i \to 0$. This completes the proof. ■

Remark 6. Note that the control law (57) is in distributed form, but the design of estimator (37) needs the global information Ω_{ff}. Thus, the design of our optimal control algorithm (59) is not completely distributed, while the implementation stage is fully distributed.

Remark 7. In this section, although the optimal performance index can be freely selected, the optimal control law (57) needs the estimator (37) to estimate the target states as compared to the control law (19) in Section 3. Therefore, the inverse optimal control is communication-free and has its own advantages as compared to the optimal control approach.

5 | SIMULATION

In this section, two simulation examples are provided to illustrate the effectiveness of the presented approaches. Consider the MASs with the communication graph topology shown in Figure 2. The shaded nodes 1, 2, and 3 denote the leaders and all the other nodes are followers. The equilibrium stress is plotted on each edge. Note that the stress is normalized so that its norm is one and the stress matrix is positive semidefinite according to Reference 16.

5.1 | Simulation of inverse optimal affine control law

To guarantee that the target formation can be achieved, the pair (A, T) is required to be controllable which guarantees that (21) has a unique solution. Note that A is Hurwitz to make sure that leaders can converge to the target positions.

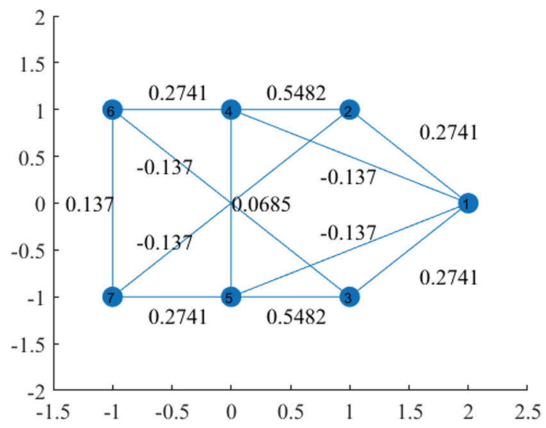

FIGURE 2 Communication graph

Therefore, the dynamic of followers in (14) is selected as

$$A = \begin{bmatrix} -1 & -0.5 \\ -2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}.$$

Assume that the weight matrices Q_2, R_1, R_2 and the coupling gain c which satisfies (23) are selected as

$$Q_2 = \begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix}, \quad R_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$R_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad c = 15.$$

Applying the theorem 1, we can get the optimal feedback gain

$$G = \begin{bmatrix} 30 & 0 & 0 & 0 \\ 0 & 30 & 0 & 0 \\ 0 & 0 & 30 & 0 \\ 0 & 0 & 0 & 30 \end{bmatrix} \otimes \begin{bmatrix} 3.06 & -0.81 & 4.39 & -2.83 \\ 0.81 & 3.06 & -2.83 & 6.89 \end{bmatrix}.$$

The simulation results of the inverse optimal control are given by Figures 3 and 4. Note that the ordinates $x_{(1)}$ and $x_{(2)}$ in Figure 3 present the first and the second element of vector x_i respectively. It is clear that the tracking error e_i approaches to the origin asymptotically and the target formation is realized.

5.2 | Simulation of optimal affine formation control law with estimator

Similarly, the system dynamic is selected as the same as the dynamic in the last simulation. The weight matrices Q_i and R_i of performance indices and the stability degree γ_i are selected as

$$Q_i = \begin{bmatrix} 100 & 0 \\ 0 & 100 \end{bmatrix}, \quad R_i = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \gamma_i = 0.1.$$

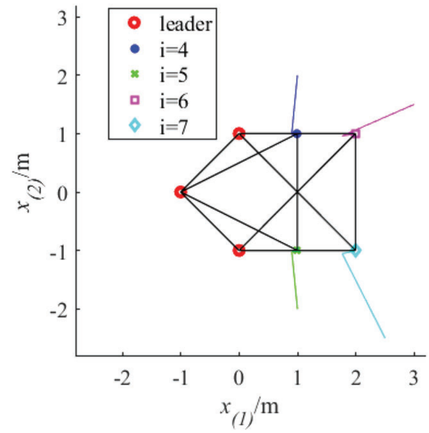

FIGURE 3 The trajectories of agents under control law (19)

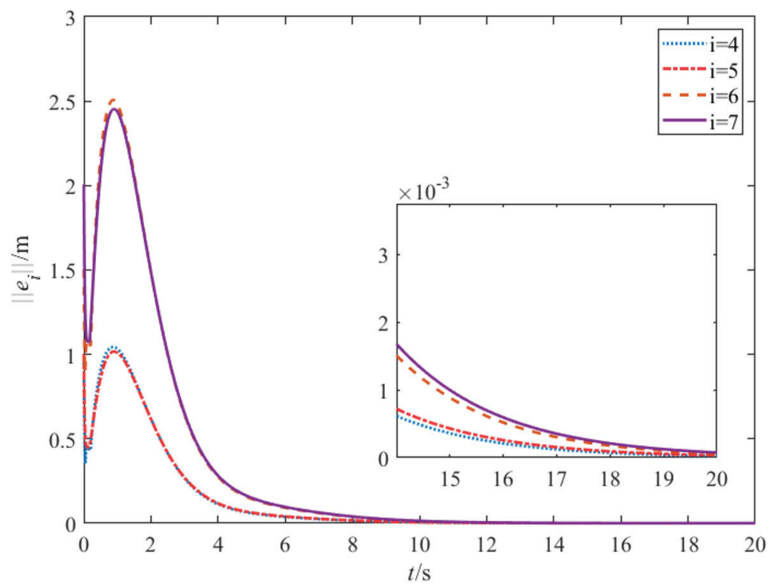

FIGURE 4 The evolution of the norm of the tracking error for each follower over time by control law (19)

Applying the result of theorem 2, we can obtain that

$$P_i^* = \begin{bmatrix} 78.2849 & -4.3656 & 19.3843 & -7.4540 \\ -4.3656 & 66.8008 & 7.3356 & 19.3109 \\ 19.3843 & 7.3356 & 10.1919 & 1.8566 \\ -7.4540 & 19.3109 & 1.8566 & 8.7719 \end{bmatrix}.$$

The results of the optimal control (57) are illustrated in Figures 5 and 6, which show that the tracking error can converge to zero asymptotically.

Then under the control law $u_i = -R_i^{-1}\overline{B}^{\mathrm{T}} P_i[z_i^{\mathrm{T}} \ \tilde{x}_i^{\mathrm{T}}]^{\mathrm{T}}$, we change the value of P_i and calculate the corresponding value of the performance index (34). The results are shown in Figure 7 and Table 1, which illustrate that the performance index (34) is optimal under the control law with $P_i = P_i^*$ because the other cases of $P_i \neq P_i^*$ give larger J_i. We also notice that when the control gain increases, the convergence time becomes shorter.

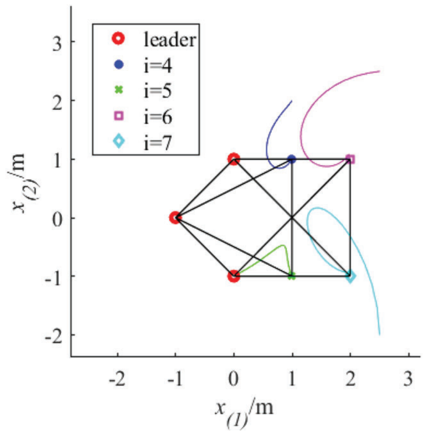

FIGURE 5 The trajectories of agents under optimal control (57)

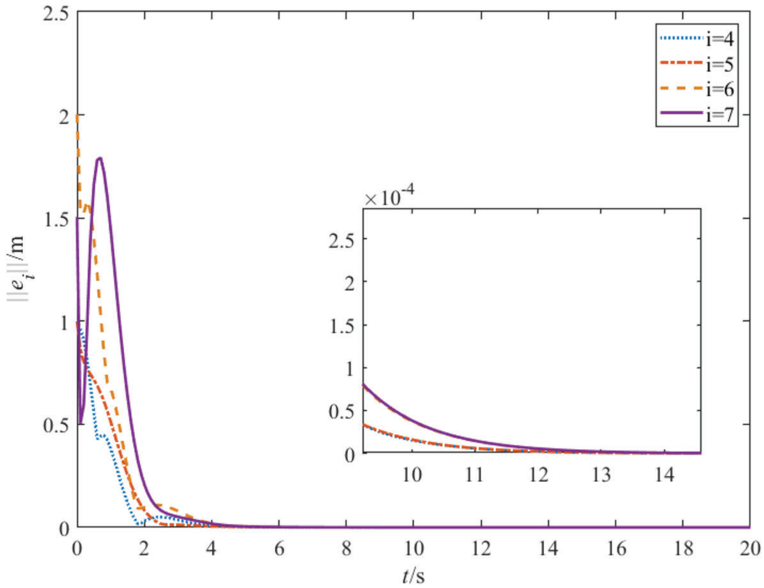

FIGURE 6 The evolution of the norm of the tracking error for each follower over time by optimal control (57)

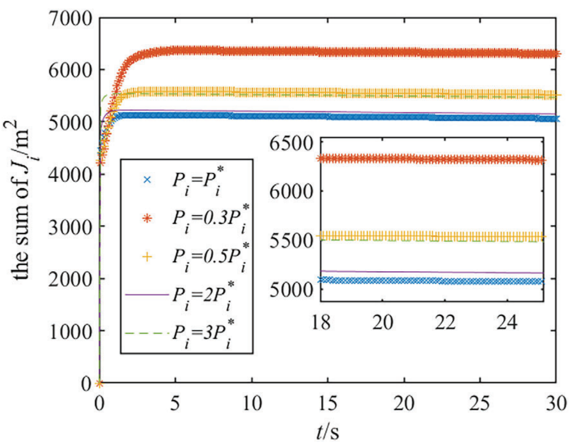

FIGURE 7 The evolution of the performance index (34) over time with different values of P_i

TABLE 1 The sum of the performance index for each follower with different P_i

P_i	$0.3P_i^*$	$0.5P_i^*$	P_i^*	$2P_i^*$	$3P_i^*$
$\sum_{i=N+1}^{N+M} J_i$	6304.34	5517.09	5061.30	5149.71	5468.32

6 | CONCLUSION

In this article, the optimal affine formation control laws have been designed for the linear MASs. First, we have designed an inverse optimal control law which can minimize the corresponding performance index. Then, we have proposed the optimal control law with estimators to guarantee a free-selected performance index optimal. Both of these control algorithms can achieve the target formation. Finally, the simulation results have illustrated the effectiveness of the presented approaches. Future work will concentrate on the optimal affine formation control with the moving target formations under directed graphs.

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DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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